

UNIT-V

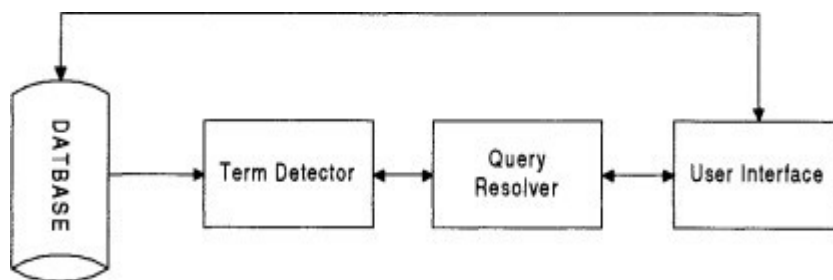
Introduction to Text Search Techniques:

The Basic Concept of a Text Scanning System is the ability for one or more users to enter Queries with the text of the Items to be Searched Sequentially and Compared to the Query Terms

When all of the text has been accessed the query is complete

One Advantage of this type Architecture is that as Soon as Item is Identified as satisfying a query the results can be presented to the user for retrieval

Text Streaming Search System Architecture



- The database contains the full text of the Items
- The Term Detector is the Special Hardware/Software that contains all of the search terms and in Some Systems the logic between the terms
- The outputs to the query resolver the detected terms
- The query Resolver performs two major functions accepting Search Statements from the users and Extracting the logic and search terms to pass to the detector
- It also accepts results from the detector and determines which queries are satisfied by the Item and possible the relevance weight Associated with Hit
- The query Resolver passes Information to the User Interface, allowing it to Continually update the search Status & on Request Retrieve any items that satisfy the user search statement
- The text streaming process is focused on finding at least one or all occurrences of a pattern of text in a Text Stream
- It is assumed that the same Alphabet is used in both Search terms and text being Streamed

Software Text Search Algorithms:

In Software Streaming Techniques, the items to be Searched is read into memory and then the algorithm is applied

There are three Major Algorithms Associated with Software Text Search

- Brute Force Approach
- Boyer-Moore
- Knuth-Morris-Pratt

Brute Force Approach:

This approach is the Simplest String Matching Algorithm

- The idea is to try and match the search String against the Input text
- If as soon as a Mis-Match is detected in the Comparison process, shift the Input text one position & Start the comparison process over
- The Expected number of comparisons when Searching an Input text string of N Characters for a pattern of M Character is

$$N_c = c/c-1(1-1/c^m)(n-m+1)+O(1)$$

Where is the expected number of Comparisons and C is the Size of the Alphabet for the text.T

The brute-force Pattern Matching Algorithm Compares

The pattern P with the text T for each possible shift of Relative to T until either

Brute-Force pattern Matching runs in time $O(nm)$

O=Order

Order of $n*m$

-T= aaa.....ah

-P=aaah

It is an algorithm is for Brute force Approach

Boyer Moore Algorithm:

- It is a String Algorithm
- It is Significantly Enhanced as the Comparison process Started at the end of the search pattern processing right to Left versus the start of the search pattern
- The advantage is that large Jumps are Mis-Matched Character in the Input Stream the search Pattern which occurs frequently(one position to another position)
- The original Boyer-Moore Algorithm is developed for additional text search Techniques
- It was originally designed to support scanning for a single Search String
- It was Expanded to handle Multiple Search Strings on a Single Pass
- It enhanced and simplified versions of the Boyer-Moore Algorithm have been developed by May Researchers (Mollier-Nielson-84, Iyengar-80)
- The Boyer-Moore's Pattern Matching Algorithm is based on two Heuristics
- Looking-glass heuristic: Compare P with a Sub-Sequence of T Moving backwards
- Character-Jump Heuristic: when a Mis-Match occur at $T[i]=C$
- If P Contains C, Shift P to Align the last occurrence of C in P with $T[i]$ Else Shift P to align $P[0]$ with $T[i+1]$
- It is an Boyer-Moore Algorithm

Knuth-Morris Pratt Algorithm:

The Knuth Morris Pratt Algorithm made a major Improvement in Previous Algorithms

- It is an text search Algorithm
 - Even in the worst Case it works well
 - Unlike the previous Algorithms it does not Depend on the length of Input String
 - It can also work as long Strings
 - The basic Concept Behind the algorithm is that whenever a Mis Match is detected, the previous Matched Characters define the Number of Characters that can be Skipped in the Input Stream Prior to Process Again
 - Starting The Comparison
 - Position: 1 2 3 4 5 6 7 8
 - Input Stream a b d a d e f g
 - It is going to Identified as Position
- When the Mis-Match Occurs in Position 4 with a "F" in the pattern and a "b" in the Input Stream this Algorithm allows the Comparison to Jump at least the three Positions associated with recognized "abd"
- Since the Mis Match on the position could be the beginning of the search strings four Positions can not be Skipped

- To Know the Number of Positions to Jump based up on Mis Match in the Search Pattern the Search Pattern is Pre-Processed to define a number of characters to be Jumped for Each Position

P=Position

S=Search Position

I=Input Stream

P	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S	a	b	c	a	b	c	a	c	a	b	c	a	a	b	c	a
I	b	a	b	c	b	a	b	c	a	b	c	a	a	b	c	a
	↑															
mismatch in position 1 shift one position																
P	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S		a	b	c	a	b	c	a	c	a	b	c	a	a	b	c
I	b	a	b	c	b	a	b	c	a	b	c	a	a	b	c	a
					↑											
mismatch in position 5, no repeat pattern, skip 3 places																
P	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S					a	b	c	a	b	c	a	c	a	b	c	a
I	b	a	b	c	b	a	b	c	a	b	c	a	a	b	c	a
					↑											
mismatch in position 5, shift one position																
P	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S						a	b	c	a	b	c	a	c	a	b	c
I	b	a	b	c	b	a	b	c	a	b	c	a	a	b	c	a
													↑			
mismatch in position 13, longest repeating pattern is "a b c a" thus skip 3																
P	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
S									a	b	c	a	b	c	a	b
I	b	a	b	c	b	a	b	c	a	b	c	a	a	b	c	a
alignment after last shift																

Hardware Text Search Systems:

The Software text Search is applicable for many Situations but faced Some Restrictions to handle Many Search Terms Simultaneously against the same text and Limits due to I/O Speeds

- One approach is to have specialized hardware Machine to perform the searches & Pass the results to the Main Computer which supported the user Interface & Retrieval of Hits
- Since the searcher is hardware based Scalability is achieved by Increasing the number of hardware Search devices
- The only Limit on speed is the time it takes to flow the text off of Secondary Storage(Disk drives) to the Searchers
- By having one search Machine Per Disk, the maximum time it takes to search a Database of any size will be the time to search one disk
- In some systems, The Disks were formatted to optimize the data flow off the drives
- Another Major Advantage of Using a hardware Text Search unit is in the Elimination of the Index that represents the Document Database
- Typically the Indexes are 70% the size of the Actual Items

- Other Advantages are that new items can be Searched as soon as received by the System rather than waiting for the Index to be Created and the search Speed is Deterministic
- Even though it may be slower than using an Index the predictability of how long it will take to stream the data provides the user with an Extract Search Time
- As hits as Discovered they can Immediately be made Available to the user versus waiting for the total search to complete as in Index Searches
- One of the earliest hardware text string Search units was the Rapid Search Machine developed by General Electric
- A More Sophisticated Search units was developed by operating Systems INC Called the Associative File Processor(AFP)
- It is Capable of Searching Against Multiple Queries at the Same Time
- In this regard we are going to discuss two types of Hardware Text Search Systems
 - GESCAN TEXT ARRAY PROCESSOR
 - FAST DATA FINDER

GESCAN TEXT ARRAY PROCESSOR:

In Some of the Systems, the Boolean Logic between terms is resolved in the term Detector Hardware

Eg:- GESCAN MACHINE

- The GESCAN System uses a Text Array Processor that Simultaneously Matches Many terms & Conditions against a given Text Stream

FAST DATA FINDER(FDF):-

The Fast Data Finder(FDP) is the most recent Specialized hardware text search unit Still in Use in Many Organizations

- It was Developed to search Text and has been used to Search English & Foreign Languages
- The early Fast Data Finders Considered of an Array of Programmable Text Processing cells Connected in Series a Pipeline Hardware Search Processor

MULTIMEDIA INFORMATION RETRIEVAL

Spoken Language Audio Retrieval:

The Speech Retrieval refers to the task of retrieving the specific pieces of Spoken Audio Data from Large Collection that pertain to a Query requested by a user Before discussing Methods for Speech retrieval it is Important to define the different types of speech retrieval tasks and the methods in which potential solutions to these tasks will be evaluated

- The speech Information Retrieval Applications the basic Scenario assumes that a user will provide a query and the system will return a list of rank ordered Documents
- It will be treat as voice of query
- It will be assigning as Ranking based Algorithm, How many no of Users given particular Query through voice or Text, no of Users are demanding particular Query , visited as site of Query related Information
- The query is generally assumed to be in the form of a string of Text-based words
- When the task requires the retrieval of Utterance Retrieval
- In this case, The purpose of SUR(Spoken Utterance Retrieval)utterance related to the query even when Multiple Utterances related to the Query Exist within a single longer Document Applications include browsing broadcast news,Voiceemail,Teleconferences&Lectures

Non-Speech Audio Retrieval:

Non-Speech Audio one of the Categories of Audio Interface, can be described as Audio Cues that accompany Specific Events

- Audio Cues means performing of object action hearing as Audio
- These cues for centuries have given human feedback on Important Information
- With the best examples like Fire Alarm is modern day, aircraft Audio cues have given Significant Amounts of Information regarding the Action of objects
- In addition to Audio cues, Music like Instrumental audio or Music is considered Non-speech audio
- In this research Music is mentioned with regard to Industry standards &Developments, Therefore this project will be directed specifically towards Audio cues
- The Non-Speech Audio Input is the use of Non-Speech sounds Such as humming or hissing for Entering Data or Controlling the User Interface
- It will be needs for user interface ,user must be alert and Attention of the sounds like Notification, Receiving Mail ,system Updates

- He had to know indications based on performing as actions
- He had to know those system Instructions based on Sounds

Graph Retrieval:

The Graph based Information Retrieval System whose query can be Expressed as a graph of Topics/subtopics, Documents are ranked with respect to a query up on relationships among Documents, relationships among Topics/Sub topics& Relationship between query terms &Documents

- The relationship between one object to another object by using as Bar chart, Pie-chart like Pictorial Representation
- User should understandable very easy way
- The system is evaluated and compared with two Information retrieval Systems on two standard text collections
- It is also Ranking based Algorithm
- Example like No of products sales, No of products refunded, Students No of pass percentage of students &No of Fail Percentage of Students it will be done as Result Analysis
- The results show that the proposed approach outperforms the other systems
- The goal of Information retrieval is to effectively retrieve Documents relevant to users queries
- The graph based approach to information retrieval, its computation is fast &Scalable and its Structure is flexible to Incorporate
- Many Performance Enhancement Techniques
- It most useful for Result Analysis, user will understandable very easily

Imagery Retrieval:

The Image Retrieval System is used for retrieving Images related to the user request from the Database

In the Presented Image Retrieval System

- An Image Retrieval System is a computer System used for browsing, Searching & Retrieving Images from a large Database of Digital Images
- The most Traditional and common methods of Image retrieval Utilize Some method of Adding Meta Data, Such Captioning, Keywords, Title or descriptions to the Images So, that retrieval can be Performed over the Annotation Words
- The Manual Image Annotation is time-Consuming Laborious and Expensive to address this, There has been a large Amount of research done on Automatic Image Annotation
- Additionally, the Increase in social web applications and the Semantic Web have Inspired , The development of several web-based Image Annotation Tools
- The need to retrieve a desired Image from a collection and to efficiently access the Information is Shared by Many groups Including Journalists, Engineers. Designers, Artists, Advertising Agencies
- Image needs and uses across users in these groups vary considerably users many require Access to Image based on Primitive Features Such as Color, Texture or Shape or users may Require access to Images based on Abstract Concepts and Symbolic Imagery
- Content Based Image Retrieval(CBIR)Technology is now beginning to Move out of the Laboratory into the marketplace
- The technology still lacks Maturity and is not yet being used in a Significant Scale

Video Retrieval:

The video is an Electronic medium for the recording Copying, Playback, Broadcasting and Display of Moving Visual Media

- Video was first developed for Mechanical Television Systems, which were Quickly replaced by CRT(Cathod-Ray-Tube)Systems which in turn were replaced by Flat Panel display
- It is Interactive web based Application which takes Video Frame from users and retrieve the information from the Database
- The Database consists of various Video data like still Video Frames, Audio &Video
- During Recent years, Methods have been developed for retrieval of videos based on their Visual features Color, Texture, Shape, Motion &Spatial-Temporal Composition are the most Common Visual Features used in Similarity Match
- Additionally ,the increase in Social Media web applications and the Semantic web
- The videos is shared by many groups designers, as well as Social Media Platforms like Youtube,...etc